A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 29

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Local people had used the area for fuelwood, timber for house building, clearings for farming, hunting and the collection of medicines and fibres, all of which protection could have stopped **1** Local people are employed as rangers, forestry workers and in ecotourism, so the forest is worth more to them standing than cleared **1** Ecotourist groups eat in local restaurants, are fed on local produce and buy locally made souvenirs, spreading the income through the community **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Trees are only felled where other trees of the same species stand upwind of the clearing, so their seed will blow in and recolonise it **1** Small trees are left standing so that they can grow and be harvested in the future when they are larger **1** Selected large trees are left because they produce many seeds that fill the gaps left by the trees that were felled **1** No mechanical or harvesting activity takes place near rivers, because aquatic ecosystems are ecologically sensitive **1** The use of large machines is restricted because they damage the trunks and roots of the trees that are not being felled **1** Track routes are planned to avoid important trees and their roots, including rare trees and those that feed monkeys or birds, and old tracks are re-used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	4
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** The tag allows the timber to be tracked from the stump where it grew to its final retail destination, such as a DIY store in the UK **1** That chain of custody is the evidence the FSC needs that the wood really came from a sustainably managed forest and was not mixed with illegally logged timber **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
01.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Ecotourism:** visitors pay to see the forest and its wildlife, and local businesses supply them **1** Timber confiscated from illegal loggers by the rangers is sold, and the money pays for conservation work **1** Electricity companies, airlines and individuals fund an area of forest as a carbon sequestration research project, because they want to offset atmospheric carbon dioxide **1** Lower-value species such as zericote are developed for furniture, musical instruments and carving, and small branches and offcuts are given to local artisans **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.5	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** The canopy is only opened in small gaps, so the forest microclimate - shade, humidity and temperature - is maintained and shade-dependent species survive **1** The soil keeps its cover of vegetation and leaf litter, and the roots that bind it, so erosion, leaching and sedimentation of rivers are far lower **1** Seed trees and small trees remain, so the gaps are recolonised by the same species and the forest regenerates naturally without replanting **1** Habitats and food sources are left in place and the forest is not fragmented, so populations of birds and mammals are not isolated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369, 386-387; AQA spec 3.5.3.3.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B **Also accept** • **abiotic** → non-living / physical factors	1

Do not write outside the box

Question	Marking guidance	Marks
02.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Deforestation **1** Ploughing of grassland / expansion of farmland **1** Reservoir creation / flooding **1** Mineral extraction, especially open-cast mining **1** Urban expansion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B **Also accept** • **biodiversity** → the variety of life: habitat, species and genetic diversity • **endemic** → found naturally only in that one place	1
03.2	**1 mark each** 1 mark for each of the 4 points below. **1 Rainforest:** high light, warm, wet all year, stable conditions for long periods, continuous productivity, many niches (stratification) **1 Antarctica:** very low temperatures, low precipitation, extreme seasonal light, 98% ice cover so almost no terrestrial life **1** Antarctic biodiversity is marine, supported by upwelling nutrients and the summer bloom – rainforest supports over half of terrestrial species on 10% of land **1** Both are important carbon stores / climate regulators but rainforest species are specialists vulnerable to small changes – Antarctic species adapted to extreme fluctuation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **biodiversity** → the variety of life: habitat, species and genetic diversity	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 3 points below. **1 Diversity:** HEP (wet winters), solar (summer) and wind peak at different times so supply is more reliable - diverse systems resist change **1** No single source could meet demand all year – battery storage on the island grid smooths fluctuations **1** Calm, dry, dull periods still leave gaps so diesel generators provide security (fossil fuel use minimised, not yet eliminated) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3
04.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Community ownership (Isle of Eigg Trust with Scottish Wildlife Trust) managing the island for people and wildlife **1** Wood biomass heating – insulation such as straw-bale housing – some electric vehicles **1 Habitat protection and creation:** fencing rare plants from grazing, planting indigenous woodland (rowan, birch, willow) – eagles and hen harriers nesting **1** Recyclables shipped off – garden / food waste composted – low-impact food production – beach litter collection **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Small groups can organise, agree and act quickly – decisions benefit people who see the results directly **1** Resources and wastes are local and visible so feedback is immediate, unlike cities supplied from far away **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
04.4	**1 mark each** 1 mark for each of the 2 points below. **1** Renewable energy is abundant but geographically and temporally variable – a continental grid moves electricity from areas of surplus (desert solar, offshore wind) to areas of demand **1** DC grids do not need every generator synchronised in phase as AC grids do, so coordination across many countries is easier **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 384-389; AQA spec 3.6.4.	
05.1	**1 mark each** 1 mark for each of the 4 points below. **1** Negative **1** Positive **1** Negative **1** Positive **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** The response to a change acts to re-establish / return the system to the previous equilibrium **1** So the change does not become self-sustaining unless a large unusual event overwhelms the negative feedback **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** eg the possible release of methane hydrate at the end of the Permian era **1** Overwhelmed the negative feedback mechanisms, causing climate change and a mass extinction / a new equilibrium **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Soil degradation, reduced water availability and loss of natural pest predators would reduce yields (negative feedback curtailing the activity) **1** Instead of reducing the activity, farmers use fertilisers, irrigation and pesticides, which further damage the natural systems that support food production **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	
06.1	**1 mark each** 1 mark for each of the 4 points below. **1 General physiological solvent:** most biological reactions take place between substances dissolved in water **1** Transport medium in blood / sap carrying gases, sugars, amino acids, nutrients, wastes **1** High specific heat capacity moderates temperature changes – evaporation absorbs heat (cooling) **1 Anomalous expansion:** ice floats so water bodies do not freeze solid – absorbs UV protecting early marine life – provides aquatic habitats **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Anomalous** → an outlier / a result that does not fit the pattern **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty	4

Do not write outside the box

Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 4 points below. **1** From about 2.7 billion years ago photosynthetic archaea / bacteria (then algae and plants) released oxygen – at first it reacted with dissolved iron (forming iron ore deposits) then built up in the oceans and atmosphere (from about 2.45 billion years ago) **1** Oxygen in the stratosphere absorbed UV, forming ozone (O3) in a dynamic equilibrium of O, O2 and O3 **1** The ozone layer reduced UV reaching the surface so organisms could survive out of the water **1** Oxygen allowed aerobic respiration releasing more energy so complex multicellular organisms evolved – many anaerobic archaea died out **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **ozone layer** → the ozone in the stratosphere that absorbs UV **Do not credit** • answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated	4
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Photosynthesis removed CO2 (a greenhouse gas) from the atmosphere, storing carbon in sediments, carbonate rocks and fossil fuels **1** This reduced infrared retention, offsetting the increasing brightness of the Sun (about 10% per billion years) so temperatures did not rise excessively **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 7-8; AQA spec 3.1.1.1-3.1.1.2.	
07.1	**1 mark each** 1 mark for each of the 4 points below. **1** k-selected: very slow growth, does not breed until 30, few eggs, so the population cannot replace losses **1** Fish reach catchable size (20) before they breed (30) so fishing can remove the entire future breeding population **1** Deep-water aggregations easily located and caught by trawling **1** Recovery takes decades so fisheries have had to be closed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4

Do not write outside the box

Question	Marking guidance	Marks
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** Herring breed at 2 years - the same age they become catchable - so there is always a breeding population and many young **1** Cod are catchable at 2 but do not mature until 4, so heavy fishing removes most fish before they breed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Larger numbers of smaller / younger fish are caught, or bigger boats, nets and sonar increase catch effort **1** Masks the decline of the breeding population until a sudden collapse **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.4	**1 mark each** 1 mark for each of the 2 points below. **1** Neither is a target species but both are caught in demersal trawls – skate now extinct in the Baltic and critically endangered elsewhere **1** Very late maturity (skate 11 years, Greenland shark 100-150 years, living over 400) and few young mean even low by-catch mortality exceeds replacement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-342, 348-349; AQA spec 3.5.2.2.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Number all possible sampling sites (eg on a grid map) **1** Select the sites using a random number generator / table – (regular shape: grid coordinates from random numbers) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	2

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis **1** So the sample does not represent the whole area and conclusions are invalid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0) **1** Fish are ectotherms so no energy is used for thermoregulation and water supports their mass – poultry grow fast and are kept warm indoors with movement restricted **1** Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Control of movement (housing / restricted grazing) so less energy is used in muscle activity **1** Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature **1 Species selection:** choose species with low FCRs (poultry, fish) – high-energy easily digested diets – slaughter at the fastest growth stage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3

Do not write outside the box

Question	Marking guidance	Marks
09.3	**1 mark each** 1 mark for each of the 4 points below. **1** Only about 10% of the energy at one trophic level is passed to the next – most is lost as heat in respiration / metabolism or in undigested material **1** Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers) **1** Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot **1** Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **trophic level** → the feeding level in a food chain • **symbiotic** → a close relationship in which both species benefit **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
10.1	**1 mark each** 1 mark for each of the 2 points below. **1** About 10% of land is already cultivated and 25% grazed **1** Most remaining land is too dry, wet, cold, hot or steep, is covered in ice, or has no soil **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** Land becomes too expensive for local people to rent / is used to grow export crops for more affluent societies **1** So people with less money lose access to the land and food they used to produce **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** It uses natural processes to supply nutrients (legumes, recycled organic matter, soil biota), control pests and provide ecological services **1** So that pesticides and artificial fertilisers become unnecessary rather than just being banned **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Unsustainable features:** reliance on chemical pesticides (resistance, non-target impacts, bans), non-renewable rock phosphate, fossil-fuel-intensive nitrate manufacture, over-exploitation of rivers / groundwater, salinisation, loss of soil organic matter, greenhouse gases from fuel, rice and livestock, loss of species providing ecosystem services • **Pest control:** cultural methods (weeding, mulching, crop rotation, barrier crops, culling), biological control, predator habitats, polyculture / companion crops, integrated control, reduced antibiotic use • **Nutrients:** recycling organic matter, crop rotation, cultivation of legumes, conservation of soil biota, reduced artificial fertiliser especially nitrate, low tillage • **Water:** low water-use crops, reservoirs and aquifer recharge, drip irrigation rather than sprays • **Soil and carbon:** low-tillage farming to maintain organic matter, reduced machinery, renewable energy, drought-tolerant rice varieties so fields are drained earlier, high-carbohydrate ground feed for cattle reduces methane • **Biodiversity:** conserve habitats of crop wild relatives, seed banks, retain hedgerows, ditches, ponds, woodland – agri-environment schemes • **Evaluation:** growing population (perhaps 12 billion) needs more food – lower trophic level diets – obstacles - cost, need for support, gaining acceptance of new techniques – learning from past land degradation (fertile crescent, US mid-west) • **Judgement:** no single strategy sufficient – combination of ecological principles and new technology – justified conclusion **Also accept**	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **salinisation** → salt building up in the soil / soil becoming salty • **biological control** → using a natural predator, parasite or pathogen of the pest **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	
	Present agriculture relies on inputs that cannot last: chemical pesticides to which pests evolve resistance and which harm non-target species, rock phosphate that is non-renewable, nitrate fertiliser made with fossil-fuel energy, and irrigation that over-draws rivers and groundwater and salinises soil, while ploughing loses soil organic matter and livestock, rice and machinery release greenhouse gases. Each can be replaced by strategies based on ecological principles. Pests are controlled culturally by weeding, mulching, rotation, barrier crops and culling, by biological control and predator habitats, by polyculture and companion crops, and by integrated control that uses chemicals only as a last resort. Nutrients come from recycled organic matter, rotation with legumes and conserved soil biota, so artificial fertiliser, especially nitrate, can be reduced. Water is saved with low water-use crops, reservoirs and aquifer recharge and drip irrigation; soil carbon is kept by low tillage and less machinery, with renewable energy for what remains; methane falls with rice varieties that allow earlier drainage and with ground, carbohydrate-rich cattle feed. Habitats of crop wild relatives, seed banks, hedgerows, ditches, ponds and woodland and agri-environment schemes protect the biodiversity that supplies these services.	
	Food demand will rise with a population that may reach 12 billion, there is little new land, and imports can reduce food availability where it is grown, so yield cannot be ignored. Some strategies cost more or need farmers to accept new techniques, and past land degradation from the fertile crescent to the American mid-west shows what happens when they are not adopted. No single strategy suffices; a combination of ecological principles with new technology is the only route to feeding more people without exhausting the resources food depends on.	
	Weaker answer - Level 1 (2 marks)	
	To make agriculture sustainable farmers should use fewer chemicals. Organic farming does not use pesticides or artificial fertilisers. Crop rotation and manure keep the soil fertile. Farmers should not waste water. Hedges should be kept for wildlife. This would make farming more sustainable but it might produce less food.	
	What makes the difference	
	• Level 1 gives general advice; Level 3 identifies each unsustainable feature and matches it with specific strategies, showing understanding of why they work.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3's evaluation considers population growth, land limits, cost, adoption and historical evidence; Level 1's is a single clause.	
	• Note that Level 3 avoids defining organic farming as 'not using chemicals' - the textbook stresses it is the positive use of natural processes.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	
11.1	**Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Definition of sustainability (Brundtland) – population x per capita impact – dynamic equilibria, feedback, tipping points, diversity • **Principles of natural systems:** renewable low-density energy, low temperatures, material cycles, non-toxic wastes – contrast with linear human systems • **Circular economy principles:** cycling materials, renewable energy, supporting ecosystems, separation of technical and biological materials, diversity, connected systems, design for extended use and re-use, optimum production, new technologies • Application to land use, water, minerals, waste, pollution, energy, food with examples (Kalundborg, Eigg, Ol Pejeta, Rio Bravo, EU initiatives, DC grids) • **Measuring progress:** ecological footprint, biocapacity, global hectares, Earth Overshoot Day, carbon footprint, Living Planet Index • **Barriers:** economics, competition, consumer behaviour, need for international agreement, existing infrastructure • Judgement on the extent achievable	25

Do not write outside the box

Question	Marking guidance	Marks
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-389; AQA spec 3.6.	
11.2	**Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Species selection:** market, range of tolerance, controllability • **Abiotic control:** temperature, light, water, soil fertility / nutrients, aeration, salinity, pH, CO_2, topography, relief, wind - methods and examples • **Biotic control:** pest groups, cultural control, biological control (successes and failures), sterile males, pheromones, genetic resistance, pesticides (properties), antibiotics, pollinators, soil biota • **Manipulation of food species:** population density, monocultures, selective breeding, asexual reproduction, cross-breeding, AI / embryo transfer, GM – energetics - intensive vs extensive, energy subsidies, ratios, FCR, food chain losses • **Impacts:** habitat loss / change, reduced biodiversity, introduced species, pesticide and nutrient pollution, GHGs, hydrological changes, soil erosion – positive: plagioclimax habitats • Social, economic and political influences shaping methods • Sustainable strategies and evaluation	25
	– conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-337; AQA spec 3.5.1.	

Do not write outside the box